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Forecasts for the *Gaia* DR4 BH revolution

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ABSTRACT

1. SIMULATING THE GAIA DETECTABLE POPULATION OF BH-STAR BINARIES

Everyone: This is a very rough draft of the methods, I would love some opinions on it and perhaps someone also willing to check my normalisation maths :)

In order to make a prediction for the detectable population of BH-star binaries, we aim to simulate the intrinsic population, with their present day properties:

$$\theta = \{m_{\rm BH}, m_{\rm comp}, M_{G,\rm comp}, e, P, \vec{x}, \vec{v}\}, \tag{1}$$

where $m_{\rm BH}$ is the mass of the black hole, $m_{\rm comp}$ is the mass of the companion star, $M_{G,\rm comp}$ is the Gaia G-band absolute magnitude of the companion star, e is the orbital eccentricity, P is the orbital period, and $\vec{x}$ and $\vec{v}$ are the position and velocity of the binary in the Milky Way. We use `COSMIC` to generate the intrinsic properties of the binaries, and `cogsworth` to compute their Gaia G-band magnitudes, positions, and velocities, as we describe below.

1.1. *Creating a sample of BH-stars from which to draw*

BH-star binaries are intrinsically rare in the Milky Way. We expect that there are around 170,000 present in the Milky Way. Given that there's around $6\times10^{10}\,{\rm M}_\odot$ of stellar mass in the Milky Way, that implies a flat Monte Carlo sample would require one to evolve about 300,000 stellar masses of binaries to get each individual BH-star. It is therefore important design a more targeted sampling strategy. It's worth considering the reasons for the rarity of BH-star binaries, because it informs how we should sample them.

BH-star binaries with low-mass stars (BH-LMS) are rare because they actually *form rarely*. They require extreme mass ratios (for example, a $30\,{\rm M}_\odot$ BH progenitor with a $1\,{\rm M}_\odot$ companion), which are rare when assuming a flat prior on mass ratio. These binaries still often fail to produce bound BH-star systems. A tight initial binary will undergo mass transfer when the primary star fills its Roche lobe, which is almost inevitably unstable given the mass ratio of the system. These common-envelope events often fail and merge due to a failed common-envelope. Alternatively, if the binary is wide enough to avoid mass transfer, any supernova kick imparted by the BH is often enough to unbind the system.

BH-star binaries with high-mass stars (BH-HMS), on the other hand, are rare because they *die quickly*. Although it is common for a high-mass binary to have a stage in which it is a bound BH and a massive star, this is short lived because of the rapid evolution of the stellar companion. As a result, these systems must form recently in the history of the Galaxy in order to be observed today.

Given the different reasoning for their rarity, we apply two different sampling strategies. For the BH-HMSs, creating an intrinsic population can be achieved through simple Monte Carlo draws from the initial distributions. For BH-LMSs, a more complex approach is required. We employ the `STROOPWAFEL` adaptive importance sampling algorithm (F. S. Broekgaarden et al. 2019) to sample only systems that are likely to form our targets. We note that separating these populations is essential, otherwise the adaptive sampling would produce negligible gains and would be dominated by the more common BH-HMSs.

We apply the subsequent analysis to each of these populations separately, and then combine the results at the end, normalising for their relative rates.

Question: Is this there a more clever way to combine these consistently?

For the `STROOPWAFEL` sampling we assume the following priors for the initial distributions of the binaries. The primary mass, m_1, is drawn from a Kroupa IMF (P. Kroupa 2001) between $5\,{\rm M}_\odot$ and $150\,{\rm M}_\odot$. The mass ratio, $q = m_2/m_1$, is drawn from a flat distribution between 0 and 1 (T. Mazeh et al. 1992). The orbital period, P, drawn from a power-law distribution with $p(P) \propto (\log_{10}(P/{\rm days}))^{-0.55} \in [0.15, 5.5]$ (H. Sana et al. 2012). The eccentricity, e, is drawn from a power-law distribution with $p(e) \propto e^{-0.45} \in [0, 0.9]$ (H. Sana et al. 2012).

We use these priors to sample populations of BH-LMSs for each metallicity bin for which we have a `MESA` grid. We then evolve these binaries with `COSMIC-METISSE`. This gives us the formation properties of the binaries in a given metallicity bin

$$\theta_{b,Z} = \{m_{\rm BH}, P_i, m_{{\rm comp},i}, e_i, t_{\rm start}, t_{\rm end}\}, \tag{2}$$

where P_i, $m_{{\rm comp},i}$, and e_i are the initial orbital period, companion mass, and eccentricity of the binary at the formation of the BH-star binaries. $t_{\rm start}$ is the time at which the system starts its BH-star phase, and $t_{\rm end}$ is the time at which it ends. The end time is defined as when the companion star evolves into a white dwarf, neutron star, or the maximum simulation time is reached (13.7 Gyr).

1.2. *Populating the Milky Way*

We next populate a Milky Way our synthetic BH-stars. This additional level of sampling is necessary since the synthetic binaries are uniformly sampled across metallicity bins, which is not representative of the Milky Way. We use the J. L. Sanders & J. Binney (2015) model for the Milky Way, which includes distributions for the lookback time, positions, velocities. We combine this model with N. Frankel et al. (2018), which provides a relation between birth time, position, and metallicity. These models account for radial migration and the inside-out growth of the galaxy, whilst maintaining a steady-state distribution of Galactic orbits. With these models, we draw $N_g = XXX$[2] individual points from the galaxy to get

$$\theta_g = \{\tau, \vec{x}_i, \vec{v}_i, Z_g\}, \tag{3}$$

where τ is the lookback time, $\vec{x}_i$ and $\vec{v}_i$ are the position and velocity of the binary in the Milky Way, and Z_g is the metallicity that we will match to the binary metallicity bins.

For each of these points, we draw a binary from our synthetic populations (the Monte Carlo sample for BH-HMSs and the `STROOPWAFEL` sample for BH-LMSs). We choose the metallicity bin closest to Z_g and draw a binary randomly. In the BH-LMS case this random draw is weighted by the binaries' adaptive importance sampling weights.

[2] Large number here. This will need to be chosen to get a representative sample, depends on detectability I suppose.

1.3. *Identifying the detectable population*

We now have a population of binaries with their formation properties. Next we need to identify which of these binaries would still be in their BH-star phase at present day and observable by *Gaia*. First, we determine whether each binary is still in its BH-star phase, based on the drawn lookback time τ and and its lifetime. We keep any binaries that are in their BH-star phase, meaning that they meet the following condition:

$$t_{\rm start} < \tau < t_{\rm end} \implies \text{still in BH-star phase} \tag{4}$$

For these binaries, we compute their present-day properties, which may differ significantly from their formation properties as a result of the evolution of the companion star. *This could be handled three different ways as far as I can see. (1) We could just re-evolve each of the BH-star systems (starting from the BH-star formation row in the bpp) and directly compute the value (CPU-hour intensive), (2) when doing the original* ***`STROOPWAFEL`*** *run we could store a lot of bcm rows and interpolate between them as necessary (memory-intensive), (3) given we have the* ***`MESA`*** *grid, and Gaia is likely insensitive to binaries that are close enough for mass transfer, we could just use the* ***`MESA`*** *grid to get the present-day properties of the companion star (too many assumptions?).*

Question: What should we do about getting the BH-star present-day state?

With this information, we can integrate the Galactic orbits of the binaries to get their present-day positions and velocities with `cogsworth`. We also use `cogsworth` to calculate the Gaia G-band absolute magnitude of the companion star, $M_{G,{\rm comp}}$. We then draw random orientations, ω, and inclinations, ι, for each binary. At this point we have the full set of present-day properties for each binary, θ.

Using these parameters, we apply `GaiaMock` (K. El-Badry et al. 2024) to get a representative sample of the detectable *Gaia* population.

Aditya: We should eventually add more details on exactly what GaiaMock does. Also potentially use Adrian's tool instead.

1.4. *Normalising to the Milky Way rate*

Finally, we must normalise our sample to the Milky Way. We have a sample of N_g binaries drawn from the Milky Way, but this is not representative of the actual number of BH-stars in the Milky Way. Given how many intrinsic systems we would create per star forming mass, we can scale by the stellar mass of the Milky Way. This additionally allows us to consistently combine the BH-HMS and BH-LMS populations, which are sampled differently.

Specifically, the number of detectable systems, $N_{\rm det}$, can be written as

$$N_{\rm det} = \frac{M_{\rm SF,MW}}{N_g} \sum_{i=1}^{N_g} \eta(Z_i) \cdot \phi_i, \tag{5}$$

where $M_{\rm SF,MW} \approx 10^{11}\,{\rm M_\odot}$ is the total star forming mass budget of the Milky Way (see Appendix B2 of T. Wagg et al. 2022)[3], $\eta(Z_i)$ is the formation efficiency of the target population (either BH-LMSs or BH-HMSs) at the metallicity of the ith binary, and ϕ_i is a heaviside function that is 1 if the binary is detectable by *Gaia* and 0 otherwise (this also accounts for systems that are not in their BH-star phase at present day). The formation efficiency is defined as the number of target binaries formed per unit star forming mass:

$$\eta(Z) = \frac{1}{f_{\rm sample}} \frac{\sum_i^N w_i \cdot \epsilon_i}{\sum_i^N w_i (m_{1,i} + m_{2,i})}, \tag{6}$$

where ϵ_i is a heaviside function that is 1 if the ith binary is a target binary and 0 otherwise, w_i is the adaptive importance sampling weight of the ith binary. We also account for the fact that we have not sampled the full initial distributions in our sampling. Therefore, we define $f_{\rm sample}$, which is the fraction the full parameter space that we sampled, as

$$f_{\rm sample} = \frac{M_{\rm sample}}{M_{\rm full}}, \tag{7}$$

where $M_{\rm sample}$ is the mean mass per binary that we sampled, and $M_{\rm full}$ is the mean mass of a full draw from our initial distributions. We fully sampled orbital period, and eccentricity distributions. However, we didn't sample single stars, we only sampled binaries with primary masses above $m_{1,\rm min}$ (where $m_{1,\rm min} = 5\,{\rm M_\odot}$), and rejected systems with $m_2 < m_{2,\rm min}$ (where $m_{2,\rm min} = 0.08\,{\rm M_\odot}$). In Appendix A we show how to analytically compute this value and that $f_{\rm sample} \sim 0.31$ for our chosen parameters.

We use this method to calculate $N_{\rm det}$ for both the BH-HMS and BH-LMS populations, and then combine them to get the total number of detectable BH-stars in the Milky Way. These totals also allow us to combine the two populations consistently, by scaling the BH-HMS and BH-LMS populations by their relative rates.

2. VARIATIONS

The plan for variations is currently all permutations of the following parameters:

- Initial mass ratio distribution: flat, $p(q) \propto q^{-1}$
- Natal kicks: fallback-modulated, no kicks, no fallback-modulation
- Remnant mass prescriptions: C. L. Fryer et al. (2012) delayed, I. Mandel & B. Müller (2020)

Tom: Note to self: the different mass ratio distribution variation has a different $f_{\rm sample}$, so we need to recalculate that for the normalisation.

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[3] Note that this greater than the present day Milky Way stellar mass, because this accounts for the fact that stars have lost mass to stellar winds and supernovae by present day.

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APPENDIX

A. CALCULATING THE FRACTION OF MASS THAT WAS SAMPLED IN OUR SIMULATIONS

In this Section we derive the fraction of total mass of the initial distributions that were sampled in our `STROOPWAFEL` simulations of BH-star binaries, $f_{\rm sample}$. As we note in Eq. 7, this is the ratio of the average mass of the binaries that we sampled, $M_{\rm sample}$, to the average total mass for the full initial distributions, $M_{\rm full}$.

We derive these total masses using conditional probabilities. We assume primary masses are sampled from an initial mass function (IMF) $\zeta(m)$, in which a mass-dependent fraction $f_{\rm bin}(m_1)$ of primaries host a companion. Companions are assigned a uniform mass ratio $q \sim \mathcal{U}(0,1)$ with $m_2 = m_1 q$, and any binary with $m_2 < m_2^{\rm min}$ is rejected. We consider the two scenarios separately: (1) the sample population, and (2) the full population of singles and binaries.

First, it is useful to note that at fixed primary mass m_1, the rejection $m_2 = m_1 q > m_2^{\rm min}$ is equivalent to conditioning $q \sim \mathcal{U}(0,1)$ on $q > a$, where we define $a \equiv m_2^{\rm min}/m_1$. A uniform distribution conditioned on $q > a$ is simply $\mathcal{U}(a,1)$, whose mean is

$$\mathbb{E}[q \mid m_1,\ \text{valid}] = \frac{1+a}{2}. \tag{A1}$$

The probability that a binary with primary m_1 is then simply the width of the remaining uniform distribution, $1-a$, so the probability that a binary is *valid* is

$$P_{\rm valid,bin}(m_1) = 1 - \frac{m_2^{\rm min}}{m_1}. \tag{A2}$$

Hence the expected total mass of a *valid* binary at fixed m_1 is

$$\mathbb{E}[m_1 + m_2 \mid m_1] = m_1\left(1 + \frac{1+a}{2}\right) = \frac{3m_1 + m_2^{\rm min}}{2}. \tag{A3}$$

A.1. *Scenario 1: BH-star sample population*

Here we retain only binaries and impose $m_1 > m_1^{\rm min}$. Because we draw q once and discard failures, the valid sample is skewed by the validity probability (A2): a primary of mass m_1 enters the kept sample with weight $\zeta(m_1) f_{\rm bin}(m_1) P_{\rm valid,bin}(m_1)$. The average total mass per binary is given by

$$M_{\rm sample} = \int \zeta f_{\rm bin}\, P_{\rm valid,bin}\, \mathbb{E}[m_1 + m_2 \mid m_1]\, {\rm d}m_1. \tag{A4}$$

Substituting (A2) and (A3), we find

$$M_{\rm sample} = \int_{m_1^{\rm min}}^{m_1^{\rm max}} \zeta(m_1)\, f_{\rm bin}(m_1) \left(\frac{3}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right) {\rm d}m_1 \tag{A5}$$

The terms in parentheses here are interpretable: $\frac{3}{2}m_1$ is the mean total mass for an unconstrained uniform-q binary (since $\langle q \rangle = \frac{1}{2}$), while the remaining terms subtract the contribution of the truncated low-q tail. Note that a *constant* $f_{\rm bin}$ cancels between numerator and denominator; the binary fraction survives in (A5) only because it is mass-dependent and thereby reshapes the primary-mass distribution of the binary subpopulation. Additionally setting $m_2^{\rm min} \to 0$ recovers $M_{\rm sample} = \int \zeta(m) m (1 + 0.5)\, {\rm d}m$.

A.2. *Scenario 2: full population*

Here we retain both singles and binaries and integrate down to the bottom of the IMF ($m_1 > m_1^{\rm min,IMF}$). Per drawn primary, the system is a single with probability $1 - f_{\rm bin}(m_1)$ (mass m_1, always kept) or a binary with probability $f_{\rm bin}(m_1)$ (mass $m_1 + m_2$, kept with probability $P_{\rm surv}$). Combining both channels, the expected retained star forming mass contribution per drawn primary is

$$\mathbb{E}[m_1 + m_2] = (1 - f_{\rm bin})m_1 + f_{\rm bin}\left(\frac{3}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right) \tag{A6}$$

$$= m_1 + f_{\rm bin}(m_1)\left(\frac{1}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right), \tag{A7}$$

where the primary mass m_1 factors out because every system contributes its primary regardless of channel, leaving $f_{\rm bin}$ to weight only the *net* extra mass from binarity, which is reduced by the low-q rejection.

The mean total mass per drawn primary is then simply the integral of (A7) over the IMF.

$$M_{\rm full} = \int_{m_1^{\rm min,IMF}}^{m_1^{\rm max}} \zeta(m_1)\left[m_1 + f_{\rm bin}(m_1)\left(\frac{1}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right)\right] {\rm d}m_1 \tag{A8}$$

Similar to above, this simplifies significantly for a constant binary fraction and $m_2^{\rm min} \to 0$, yielding

$$M_{\rm full,simple} = \int_{m_1^{\rm min,IMF}}^{m_1^{\rm max}} \zeta(m_1)\left[m_1(1 + f_{\rm bin}/2)\right] {\rm d}m_1 \tag{A9}$$

A.3. *Evaluation using our assumptions*

We retrieve an expression for $f_{\rm sample}$ by combining Eqs. (A5) and (A8). The integrals can be evaluated numerically for any choice of IMF $\zeta(m)$, binary fraction $f_{\rm bin}(m)$, and mass limits.

$$f_{\rm sample} = \frac{\displaystyle\int_{m_1^{\rm min}}^{m_1^{\rm max}} \zeta(m_1)\, f_{\rm bin}(m_1)\left(\frac{3}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right){\rm d}m_1}{\displaystyle\int_{m_1^{\rm min,IMF}}^{m_1^{\rm max}} \zeta(m_1)\left[m_1 + f_{\rm bin}(m_1)\left(\frac{1}{2}m_1 - m_2^{\rm min} - \frac{(m_2^{\rm min})^2}{2m_1}\right)\right]{\rm d}m_1} \tag{A10}$$

A more general expression for any general mass ratio distribution $p(q)$ is:

$$f_{\rm sample,general} = \frac{\displaystyle\int_{m_1^{\rm min}}^{m_1^{\rm max}} \zeta(m_1)\, f_{\rm bin}(m_1) m_1 \int_{m_2^{\rm min}/m_1}^{1} (1+q)p(q)\,{\rm d}q\,{\rm d}m_1}{\displaystyle\int_{m_1^{\rm min,IMF}}^{m_1^{\rm max}} \int_{m_2^{\rm min}/m_1}^{1} \left[f_{\rm bin}(m_1)\cdot m_1(1+q) + (1 - f_{\rm bin}(m_1))m_1\right]\zeta(m_1)p(q)\,{\rm d}q\,{\rm d}m_1} \tag{A11}$$

For these simulations, we set $m_1^{\rm min} = 5\,{\rm M}_\odot$, $m_2^{\rm min} = 0.08\,{\rm M}_\odot$, and $m_1^{\rm max} = 150\,{\rm M}_\odot$. We assume an IMF for $\zeta(m)$ from P. Kroupa (2001), and a mass-dependent binary fraction following S. S. R. Offner et al. (2023). We numerically integrated Eqs. (A5) and (A8) to find $M_{\rm sample} \approx 0.23\,{\rm M}_\odot$ and $M_{\rm full} \approx 0.76\,{\rm M}_\odot$, giving $f_{\rm sample} \approx 0.31$.